Measuring Sustainable Accessibility at the Neighbourhood scale:

a GIS-based Multimodal index for Access to Daily Opportunities

Thesis Defence

Master Metropolitan Analysis Design and Engineering

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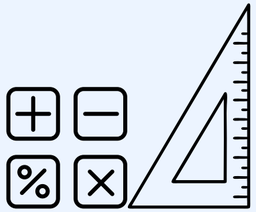
Question

Without using a car, how many of your daily needs can you realistically reach within a travel time you would still find acceptable?

Problem statement



Policy ambition



Not yet appropriate measure

Definition

Sustainable Accessibility

The ease with which individuals can reach essential opportunities for daily life within acceptable travel times using only **sustainable transport modes**

- **Essential opportunities:** What destinations are necessary to reach in daily life?
- **Acceptable travel times:** How far are people willing to travel to specific destinations?
- **Sustainable transport modes:** Travelling without using private car

Main Research Question

How can sustainable accessibility to daily opportunities be measured at the neighbourhood level using multimodal sustainable transport?

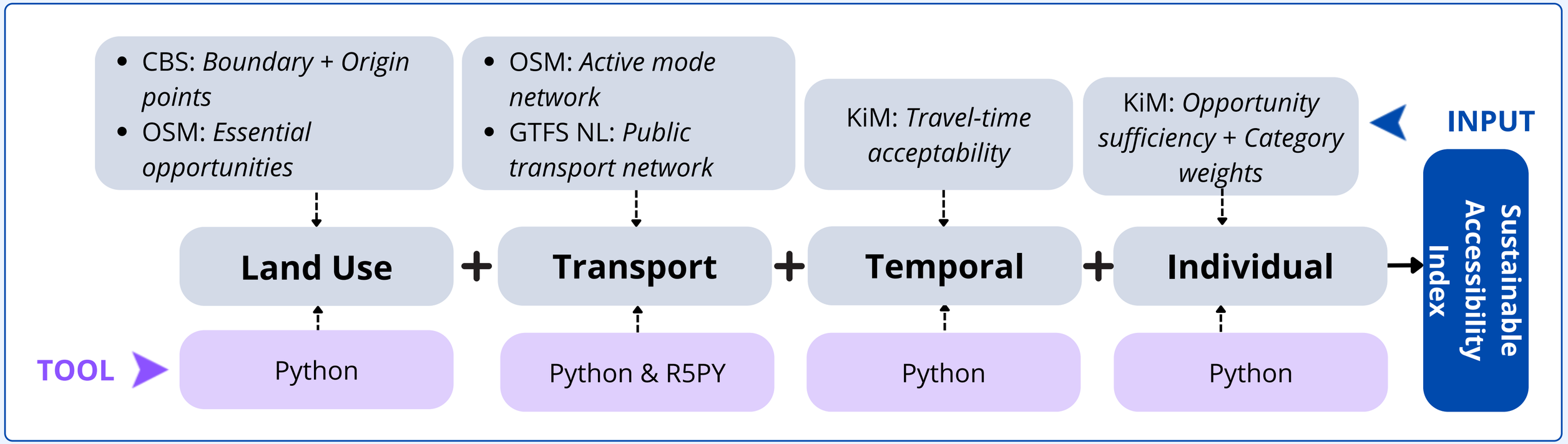
Approach

Sustainable Accessibility Index

Components of Sustainable Accessibility

- Land use → Distribution of opportunities
- Transport → Sustainable transport modes
- Temporal → Travel time
- Individual → Baseline-user and Opportunity sufficiency

Introduction Sustainable Accessibility Index



Goal

Comparing Sustainable Accessibility between neighbourhoods

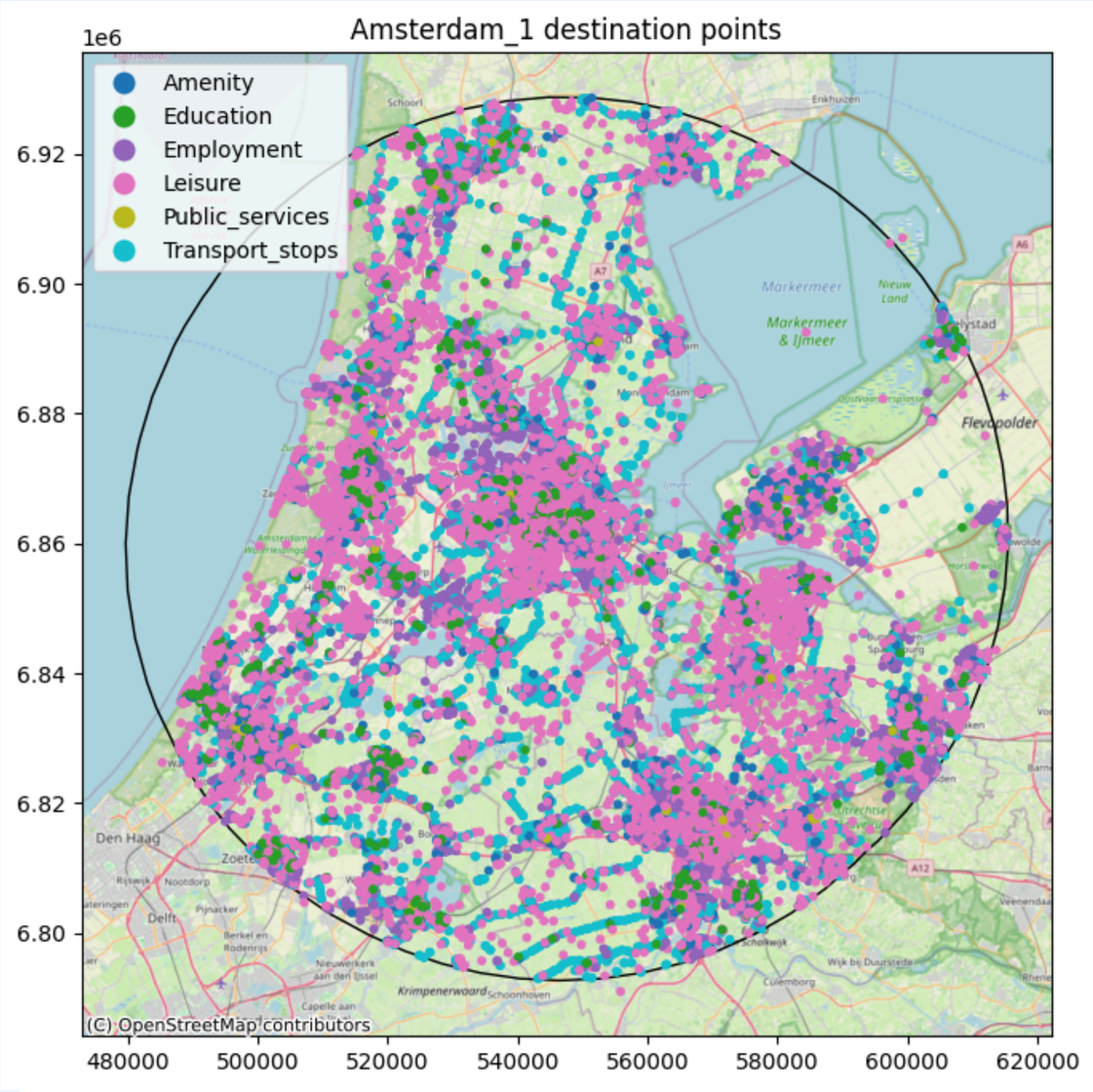
SA Index Calculation Steps

1. *Preprocessing*

Study area **Bufferzone (40km)*

Origin points

OSM destinations



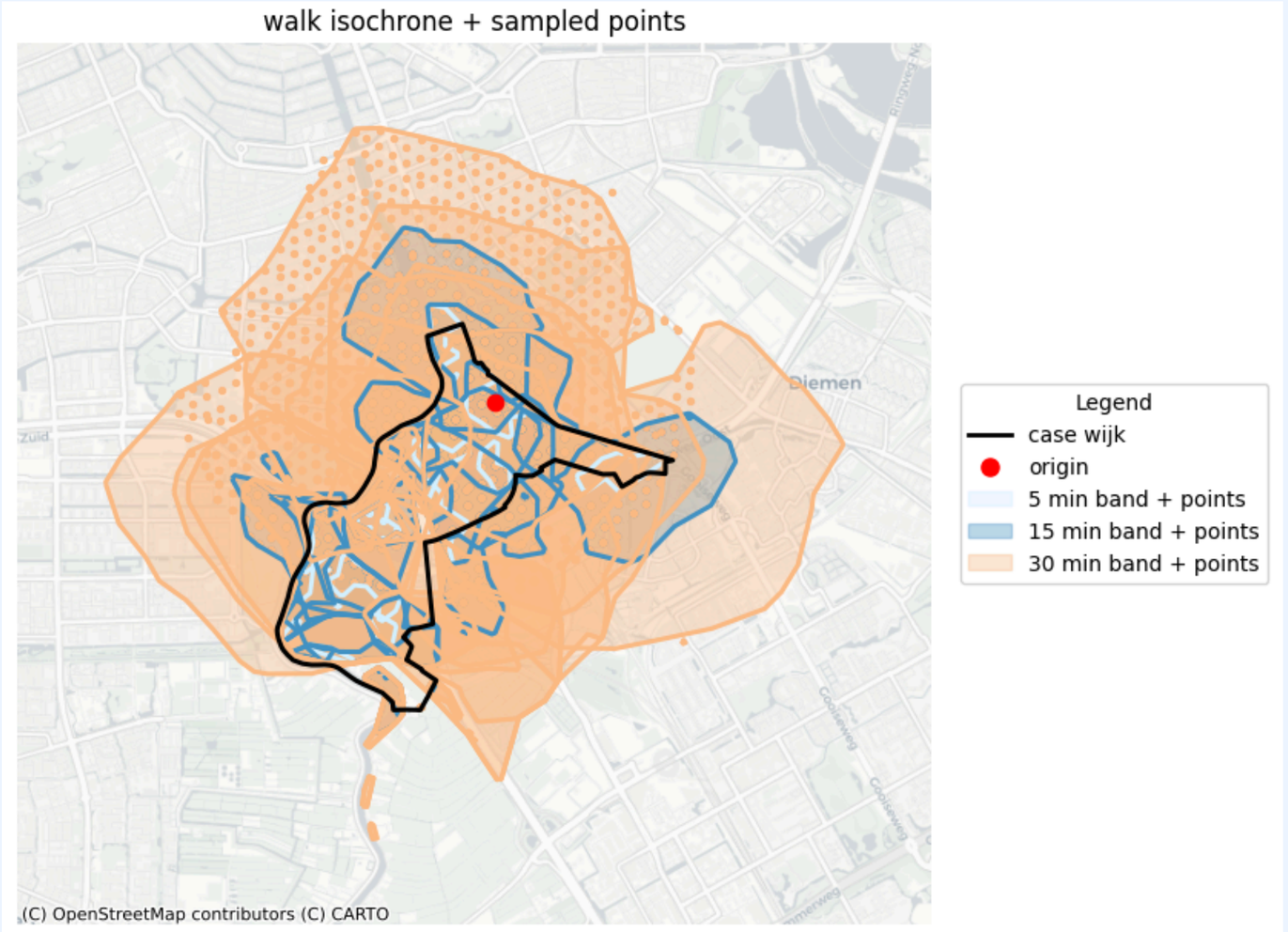
SA Index Calculation Steps

Origin points

OSM destinations

2. *Calculation travel times*

Routable network → Isochrones per band-mode-origin combo



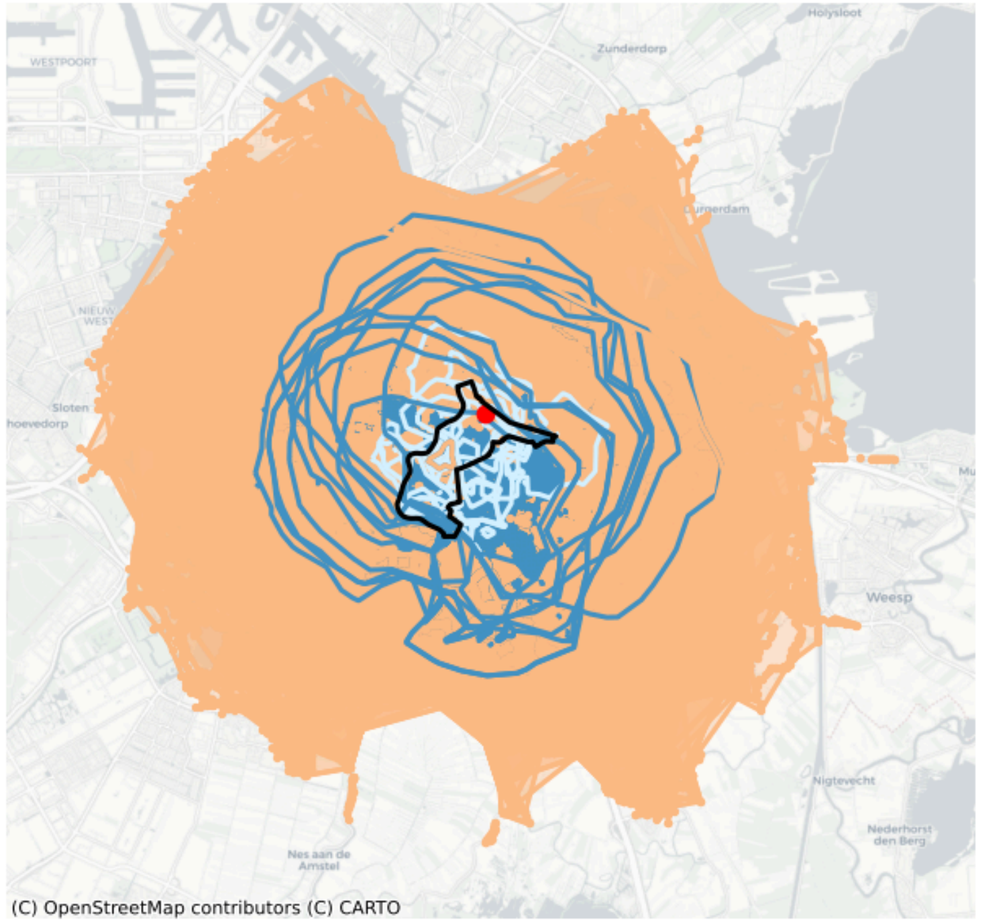
SA Index Calculation Steps

Origin points

OSM destinations

2. *Calculation travel times*

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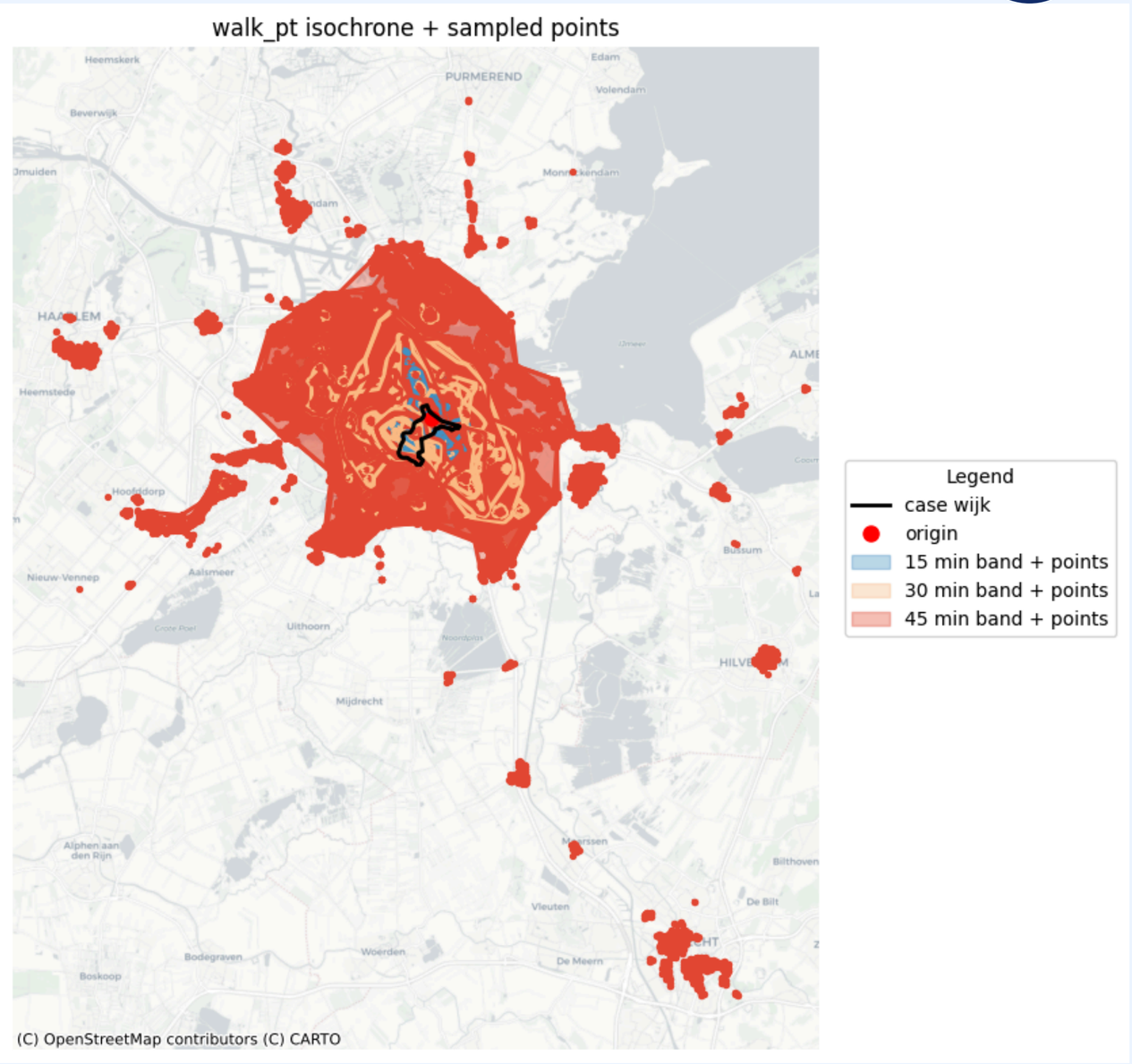
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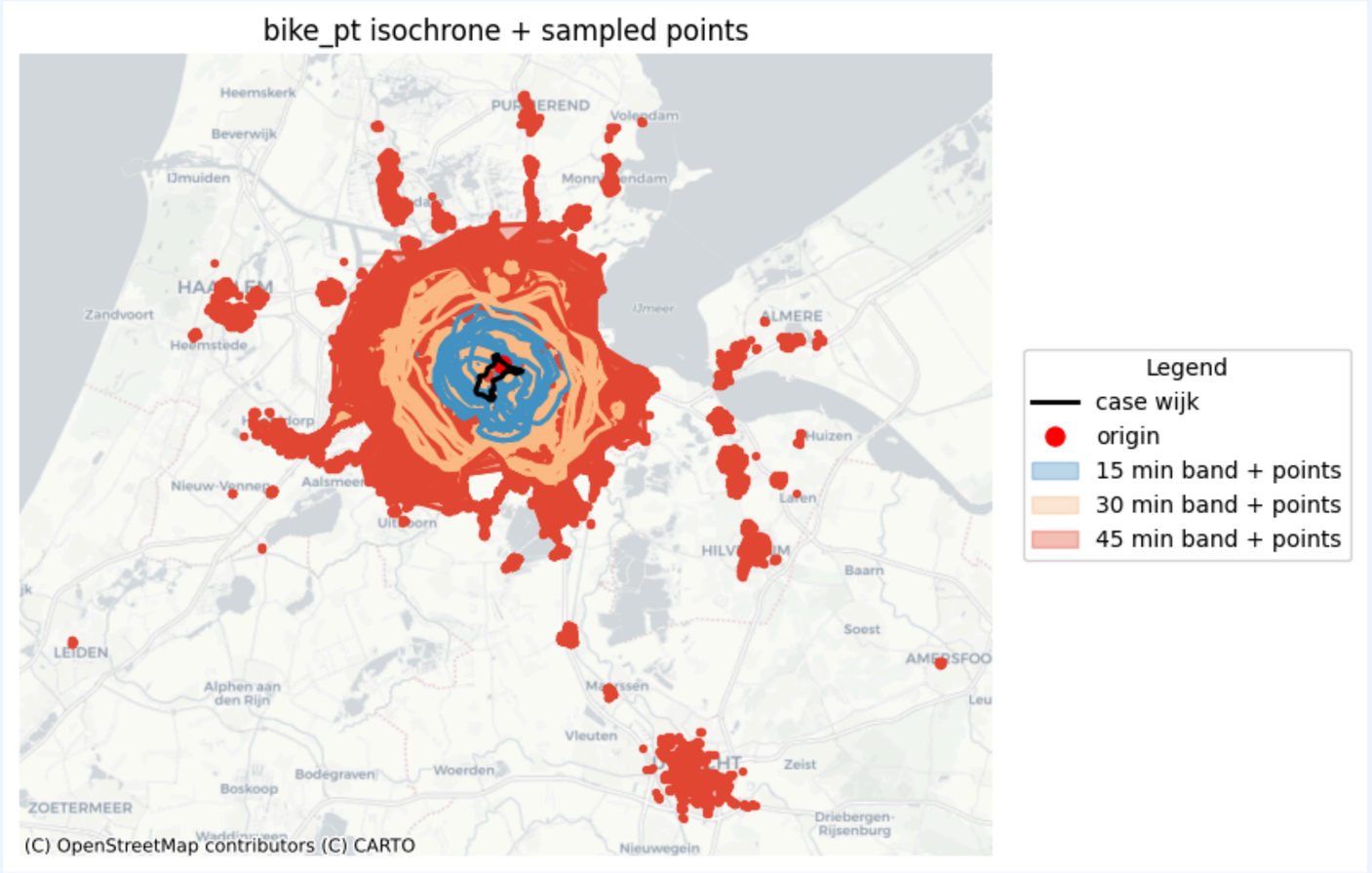
SA Index Calculation Steps

Origin points

OSM destinations

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SA Index Calculation Steps

Origin points

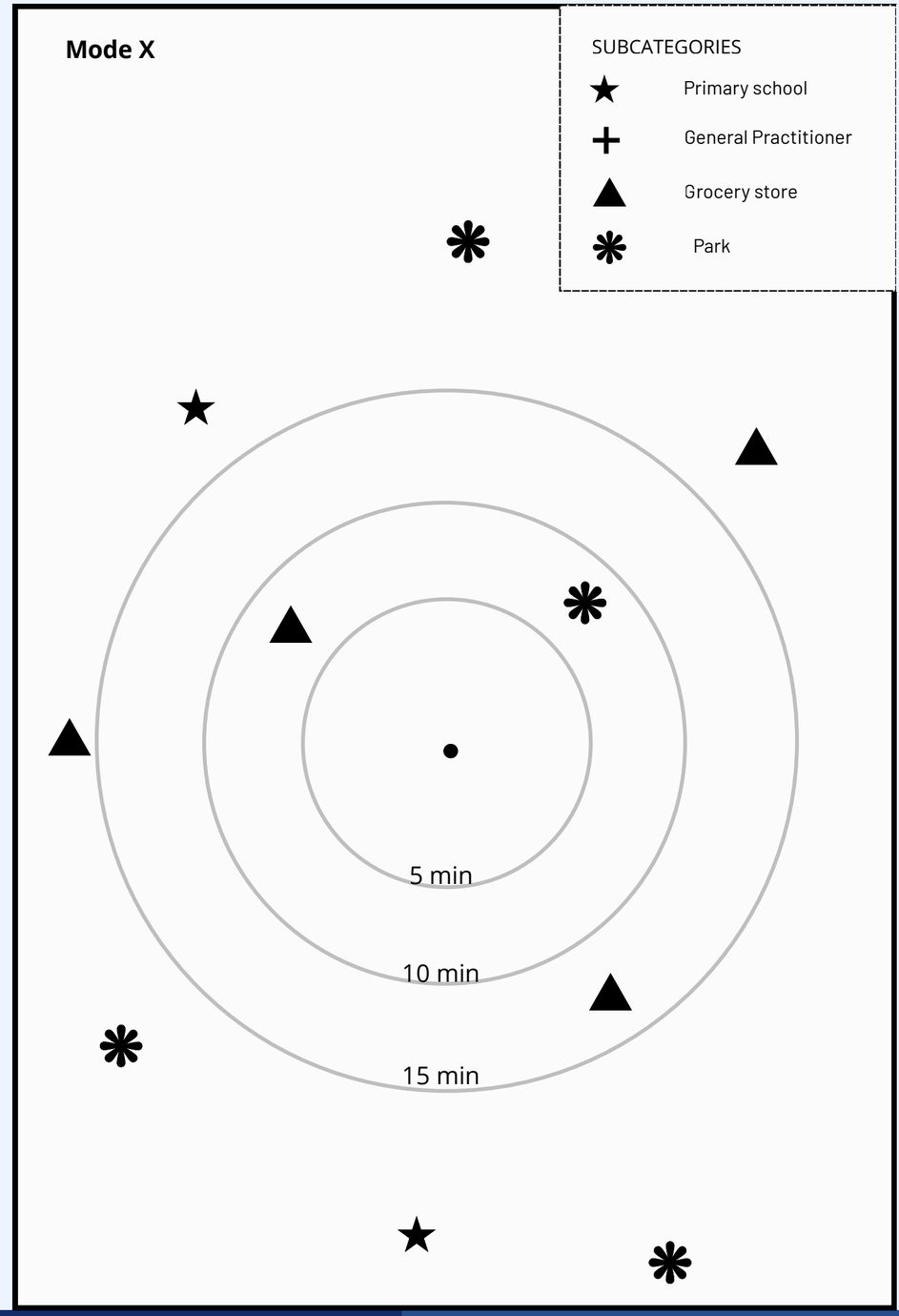
OSM destinations

Routable network

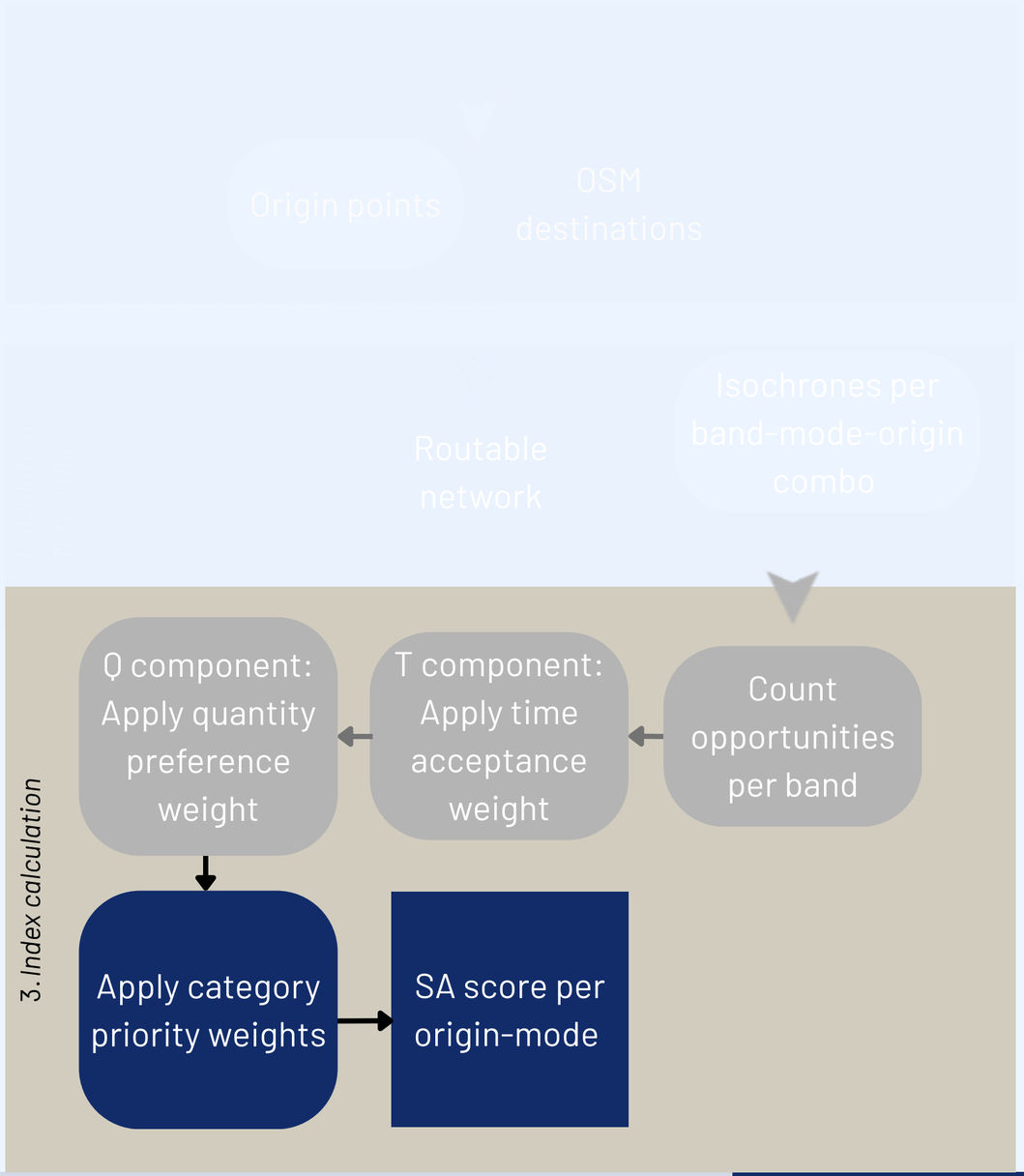
Isochrones per band-mode-origin combo

3. *Index calculation*

Count opportunities per band → T component score → Q component score



SA Index Calculation Steps



	origin_id	mode	origin_all_category_score
0	origin_00000	bike	0.842646
1	origin_00000	bike_pt	0.838600
2	origin_00000	walk	0.426411
3	origin_00000	walk_pt	0.776883
4	origin_00001	bike	0.426335
5	origin_00001	bike_pt	0.778303
6	origin_00001	walk	0.024244
7	origin_00001	walk_pt	0.198516
8	origin_00002	bike	0.500926
9	origin_00002	bike_pt	0.797649

SA Index Calculation Steps

Origin points

OSM destinations

Routable network

Isochrones per band-mode-origin combo

3. *Index calculation*

Count opportunities per band

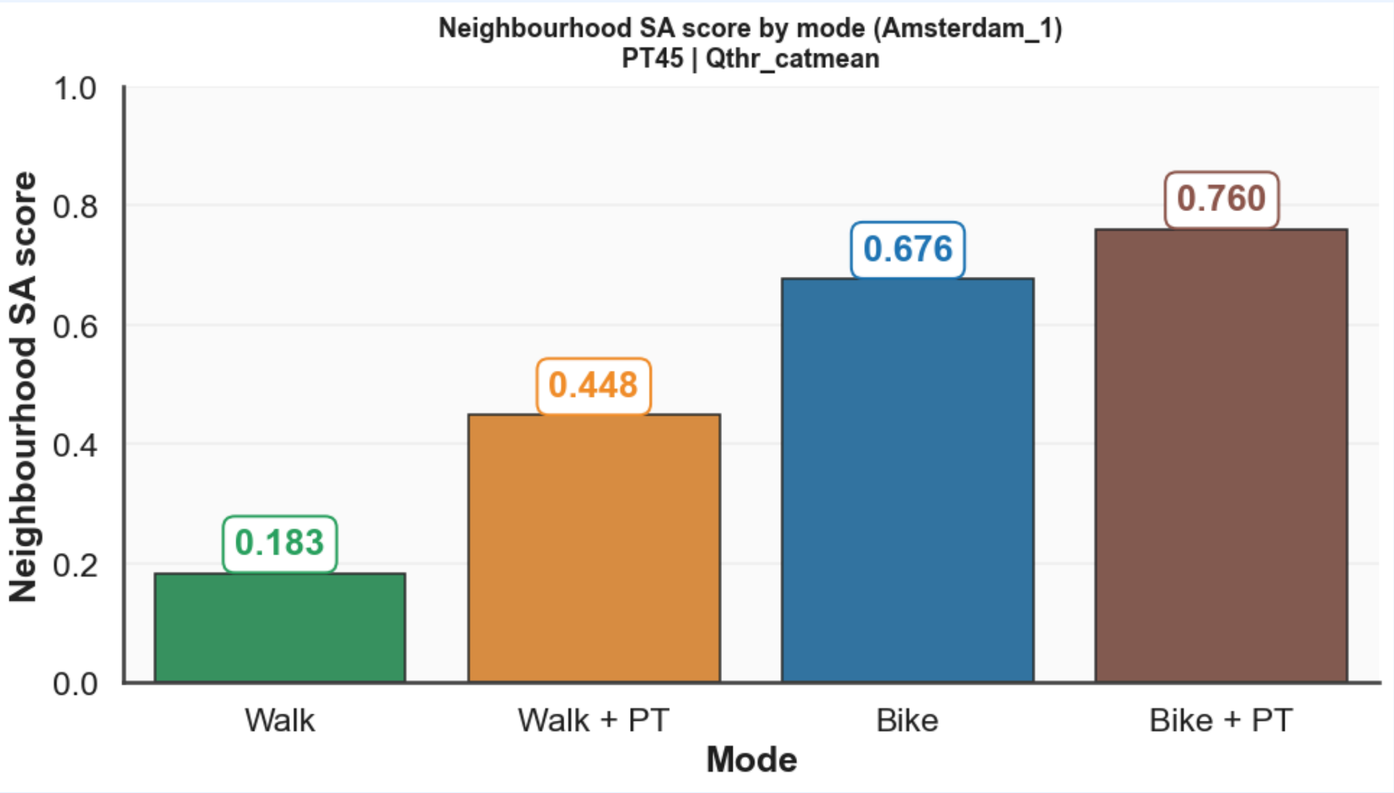
T component: Apply time acceptance weight

Q component: Apply quantity preference weight

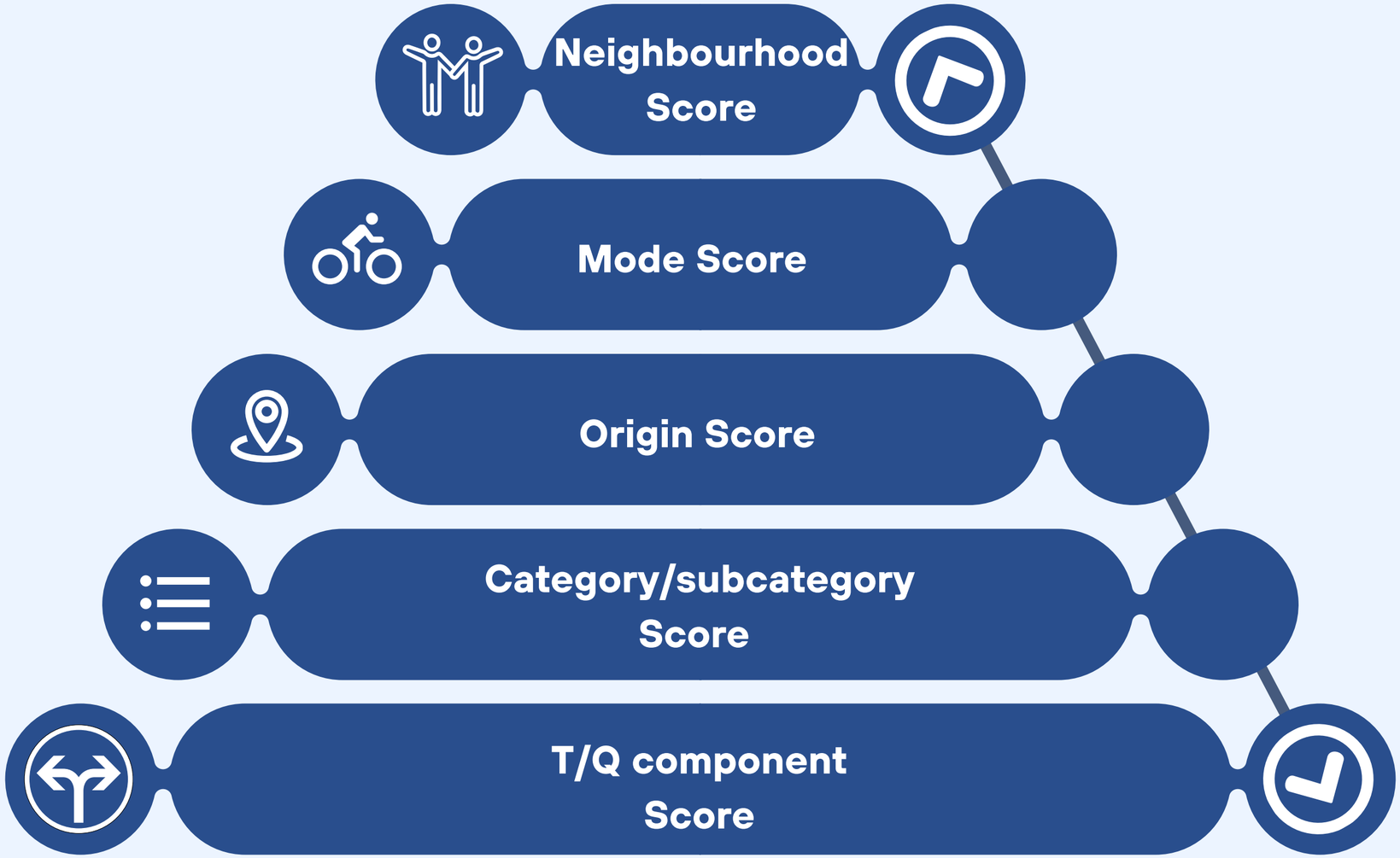
Apply category priority weights

SA score per origin-mode

SA score neighbourhood



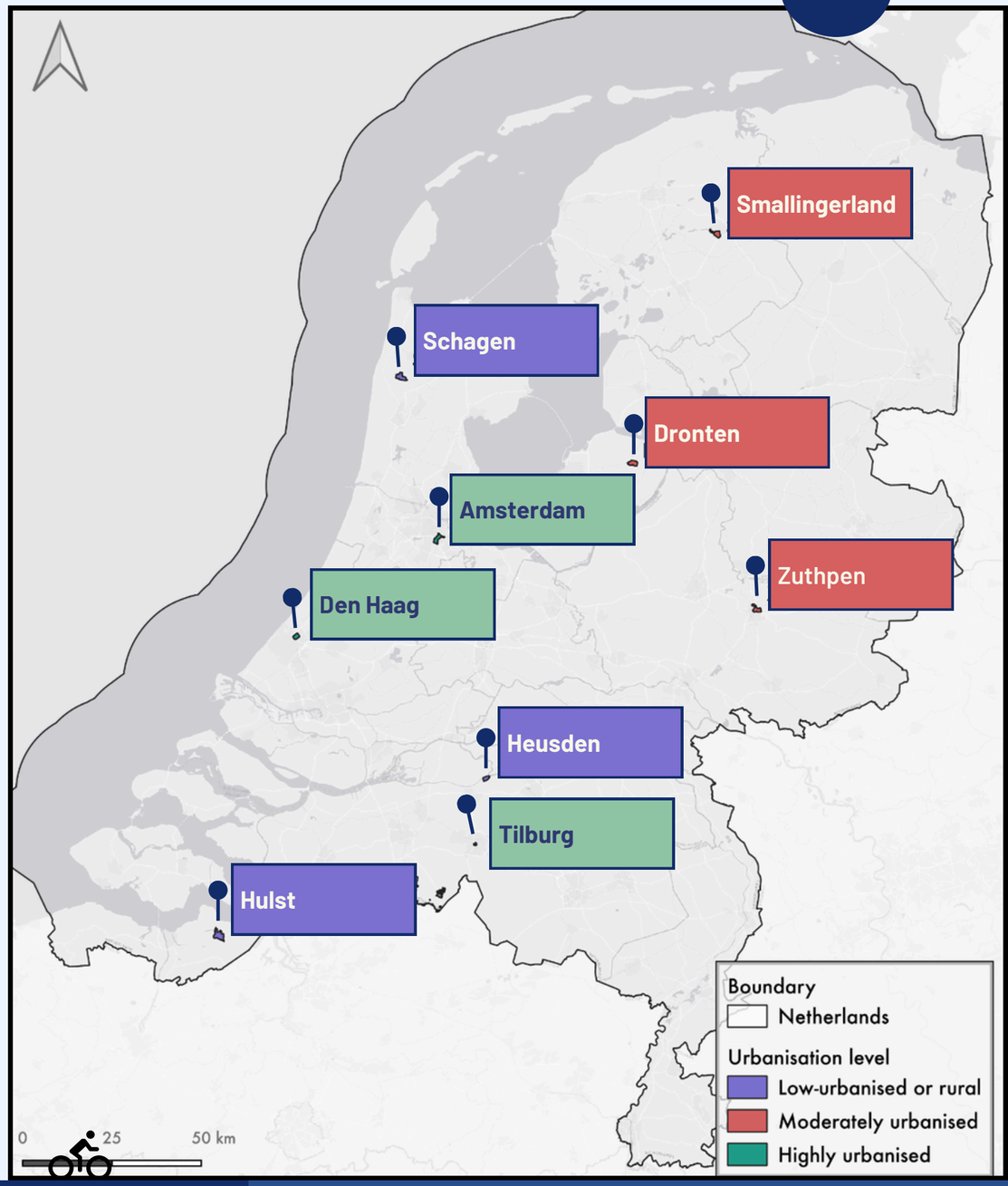
Summary SAI


Neighbourhood Score
Mode Score
Origin Score
Category/subcategory Score
T/Q component Score

Case Study Selection

9 case study areas selected based on Degree of Urbanisation

Urbanisation degree and colour	Selected Municipalities	Selected Neighbourhoods
1: Highly urbanised	Amsterdam, The Hague, Tilburg	Omval/Overamstel, Bezuidenhout, Bouwmeester
3: Moderately urbanised	Zutphen, Dronten, Smallingerland	Noordveen, Dronten Midden, De Drait
5: Low-urbanised or rural	Heusden, Hulst, Schagen	Heesbeen, Kuitaart, Tuitjenhorn

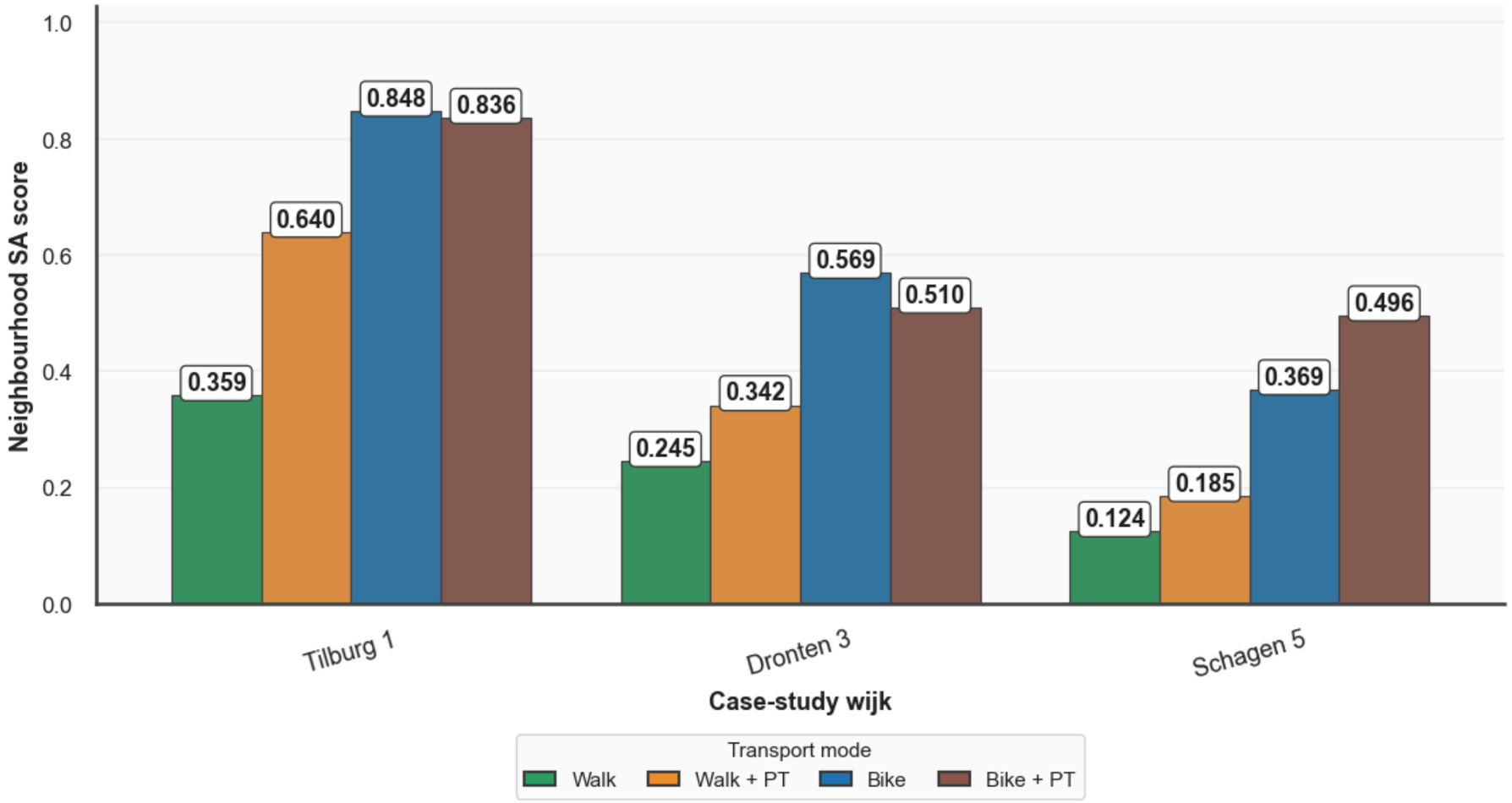


Hypotheses

It is expected that:

- H1: Higher urbanised neighbourhoods - higher sustainable accessibility
- H2: Multimodality increases sustainable accessibility
- H3: Average hide neighbourhood differences

H1: More urbanised neighbourhoods – higher sustainable accessibility



Note: within each case-study wijk, the four bars compare the four transport modes. The bar height is the neighbourhood SA score on a 0-1 scale, so higher bars mean better sustainable accessibility. White labels show the exact score, and the colors always map to the same mode across all figures.

Finding

- **Neighbourhood SA Scores follow the urbanisation gradient**

H2: Multimodality increases sustainable accessibility

Walk vs Walk + PT

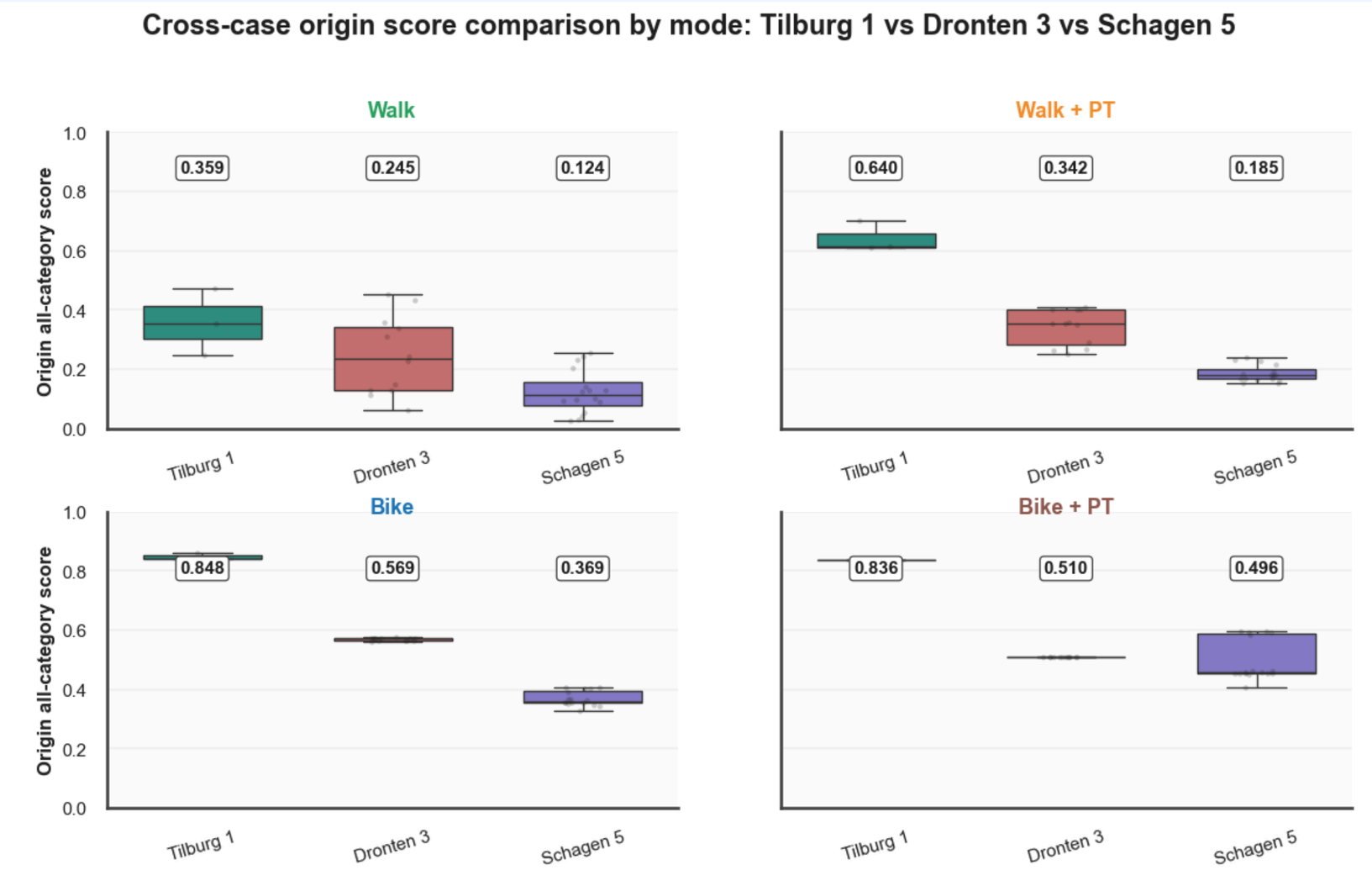
Case study	SA Score	SA Score
Tilburg	0.359	0.640
Dronten	0.245	0.342
Schagen	0.124	0.185

Bike vs Bike + PT

Case study	SA Score	SA Score
Tilburg	0.848	0.836
Dronten	0.569	0.510
Schagen	0.369	0.496

Finding: Multimodal combinations often outperform unimodal modes

H3: Averages hide neighbourhood differences



Note: Each panel shows one transport mode. The box shows the spread of origin all-category scores for the three case-study wijken, the dots are individual origins, and the white labels are the mean origin scores for each case. Case colors stay consistent across the figure, and higher values mean stronger accessibility outcomes on the 0-1 scale.

Finding

- **Higher neighbourhood SA scores can hide unequal access within the neighbourhood**

Internal Variation Amsterdam Map



Conclusion results

- Accessibility gaps can be a result of various mechanisms
- Sensitivity analysis shows robustness

The SAI is most useful as a diagnostic tool

Use of SAI

Limitations

- Data quality and completeness
- Opportunity quality and capacity
- Temporal and behavioural simplification

Recommendations for responsible use

- Specify the assumptions
- Use diagnostic layers to understand the accessibility gap
- Match interventions to accessibilty gap

Key takeaway

SAI makes Sustainable Accessibility Comparable on Neighbourhood Level and Explainable

Thank you

